Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12CE82]

EIGHTH SEMESTER B.E EXAMINATION, JUNE – 2016

ADVANCED CONCRETE TECHNOLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Any missing data can be suitably assumed.
3. Use of BIS codes 10262:2009 and IS456 : 2000 permitted.

1. a. Explain in detail about the structure of hydrated cement paste. **08**

b. Discuss the effects of superplasticizers on the properties of concrete. **06**

2. a. What parameters define rheology of fresh concrete? Explain. **08**

b. Discuss the factors affecting strength of concrete. **06**

3. Obtain the mix design for M40 grade of concrete using following data and codal procedure as per IS10262:2009 **14**

a. Design stipulation :

i.	Characteristic compressive Strength required in field for 28 days	:	40MPa
ii.	Max. size of aggregate	:	20mm (angular)
iii.	Degree of supervision	:	Good
iv.	Type of exposure	:	Mild

b. Material Test data :

i.	Specific gravity of cement	:	3.15
ii.	Specific gravity of CA	:	2.6
iii.	Specific gravity of FA	:	2.6
iv.	Water absorption: CA FA	:	 1% 1%
v.	Surface moisture: CA FA	:	 NO 2%

Grading of CA is conforming to Table 2 of IS383 and grading of FA is falling in zone II.

4. a. Define Self Compacting Concrete(SCC). Explain the necessities of obtaining SCC. **08**

b. What is FRC? Discuss the properties of various fibers and its effects on concrete. **06**

5. a. Explain the stress – strain characteristics in concrete. **07**

b. Explain the flexural strength test procedure for hardened concrete. **07**

6. Write short notes on :

i. Geo-polymer concrete **07**

ii. High Density concrete **07**

7. a. Explain the Rebound Hammer (Schmidt's) test. **07**

b. What are NDT methods? Mention the codal provisions for NDT in assessing the strength of concrete. **07**

8. a. How does curing condition influence the strength of concrete? What are the methods of curing? **08**

b. Explain the concreting practices in hot weather condition. **06**